

Introduction to VLSI & Cadence Virtuoso

ICE 3140

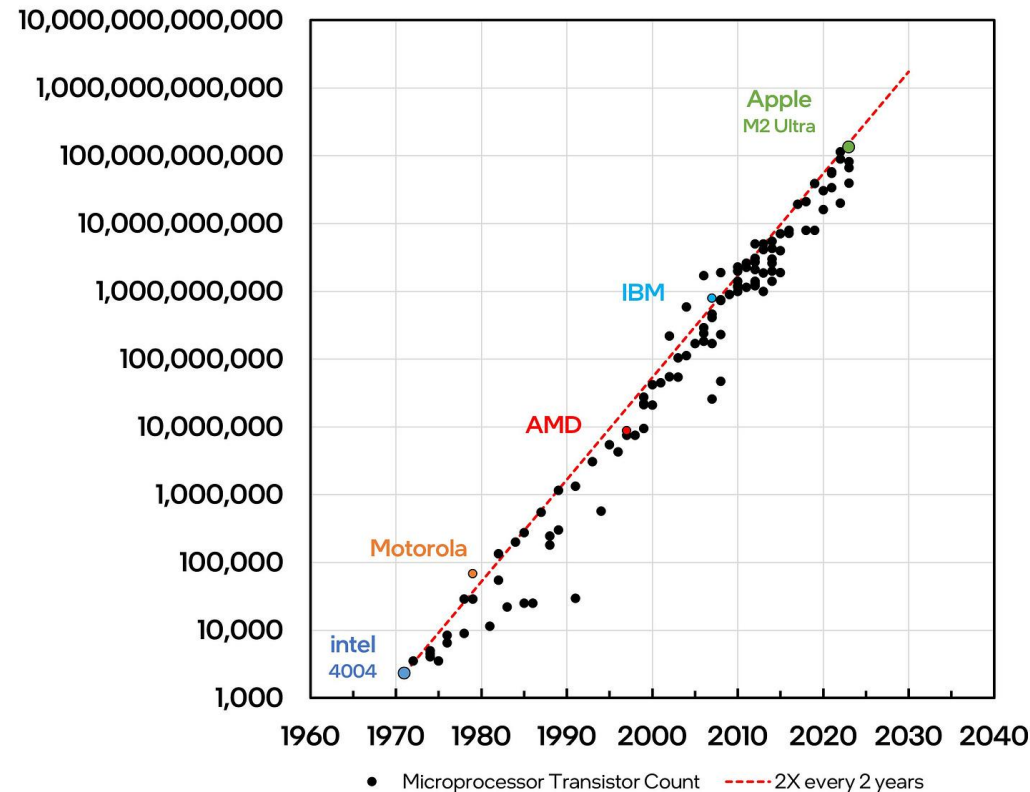
VLSI

VLSI stands for Very Large Scale Integration. This is the field which involves packing more and more logic devices into smaller and smaller areas. And it is the process of creating an integrated circuits (IC) by combining thousands of transistors into a single chip.



Moore's Law

- Moore's Law is the observation and prediction that **the number of transistors on a microchip doubles roughly every two years**, leading to exponential growth in computing power and efficiency, driven by industry innovation
- Originally an observation by Intel co-founder **Gordon Moore** in **1965**, it became a guiding principle



TYPES OF VLSI (ICS)

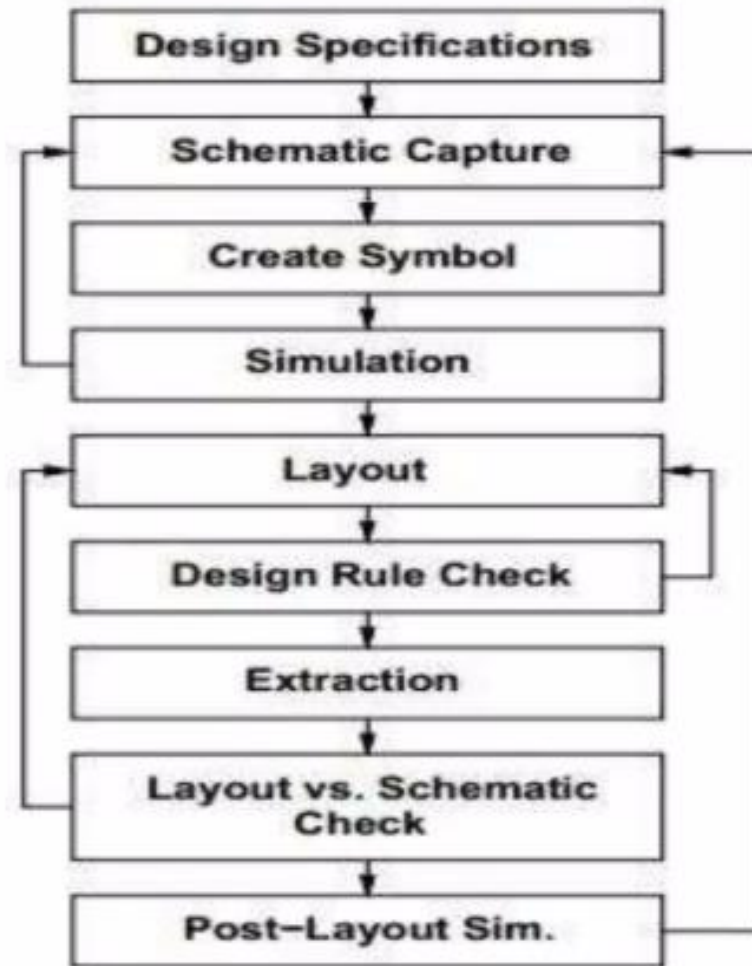
There are some type on the basis of no. of transistors:-

- SSI
- MSI
- LSI
- VLSI
- ULSI

Name	Signification	Year	Number of Transistors	Number of Logic Gates
SSI	small-scale integration	1964	1 to 10	1 to 12
MSI	medium-scale integration	1968	10 to 500	13 to 99
LSI	large-scale integration	1971	500 to 20,000	100 to 9,999
VLSI	very large-scale integration	1980	20,000 to 1,000,000	10,000 to 99,999
ULSI	ultra-large-scale integration	1984	1,000,000 and more	100,000 and more

VLSI DESIGN FLOW

A design flow of VLSI is the sequence of processes or steps involved in the making of an integrated circuit(IC).



Daily Life Applications

- Refrigerator
- Calculator
- Water purifier
- Music player
- Smart watch
- Air conditioner
- Pendrive
- Toaster
- Dish washer

ADVANTAGES

- Reduces the size of circuits.
- Reduces the cost of the devices.
- Increase operating speed of circuits.
- Less power consumption.
- High reliability.
- Occupies less area.

CMOS Technology

NMOS (N-channel)

- **Carrier Type:** Electrons (Negative).
- **Turn-on Condition:** High gate voltage ($V_{gs} > V_{th}$).
- **Substrate:** P-type.
- **Circuit Role:** Pull-down networks (connect to ground).
- **Characteristics:** Faster switching, higher power consumption, smaller size, better drive strength.

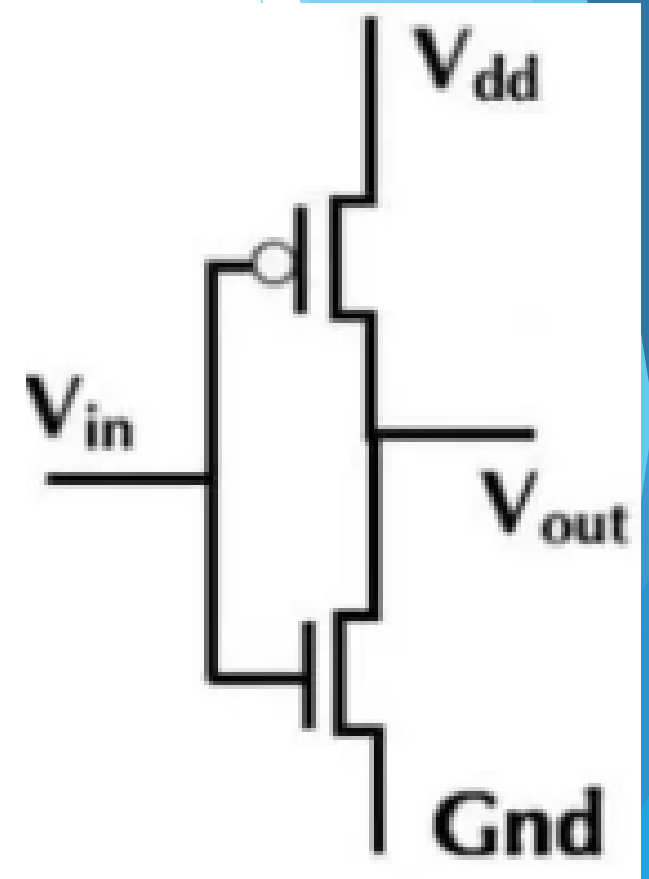
PMOS (P-channel)

- **Carrier Type:** Holes (Positive).
- **Turn-on Condition:** Low gate voltage ($V_{gs} < V_{th}$, negative).
- **Substrate:** N-type.
- **Circuit Role:** Pull-up networks (connect to VDD/positive rail).
- **Characteristics:** Slower switching, lower power consumption, better noise immunity, lower leakage.

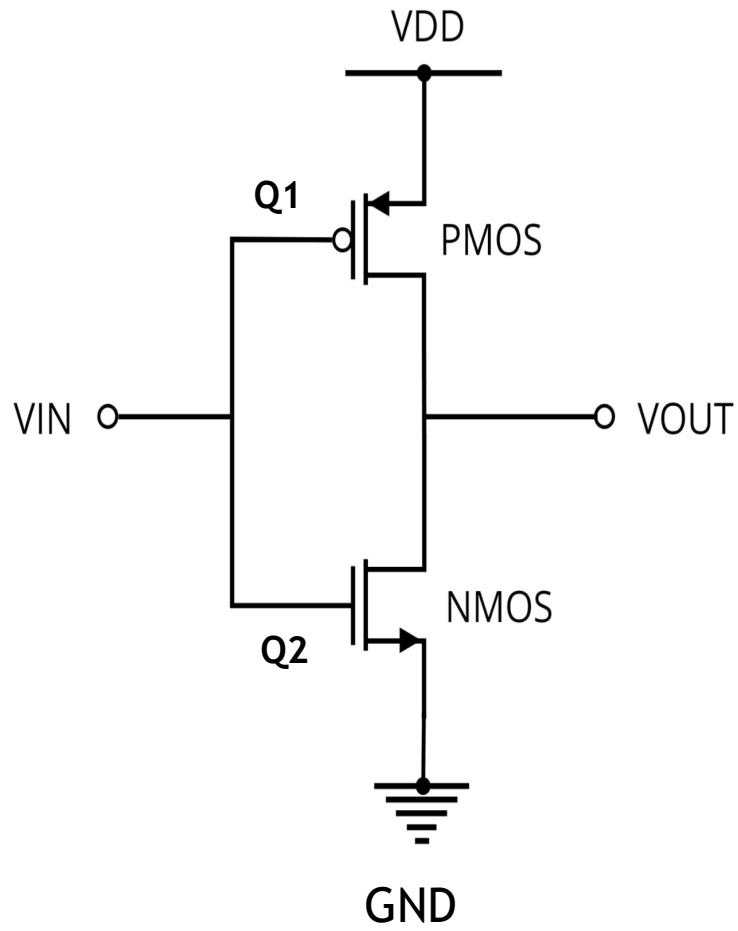
CMOS Technology in VLSI & Working:

- In CMOS technology, both N-type and P-type transistors are used to design logic functions.
- The same signal which turns ON a transistor of one type is used to turn OFF a transistor of the other type.
- This characteristic allows the design of logic devices using only simple switches, without the need for a pull-up resistor.

- Thus, if both a p-type and n-type transistor have their gates connected to the same input, the p-type MOSFET will be ON when the n-type MOSFET is OFF, and vice-versa.
- The networks are arranged such that one is ON and the other OFF for any input pattern as shown in the figure below.
- CMOS offers relatively high speed, low power dissipation, high noise margins in both states, and will operate over a wide range of source and input voltages (provided the source voltage is fixed).
- The inverter circuit as shown in the figure below. It consists of [PMOS and NMOS FET](#). The input A serves as the gate voltage for both transistors.
- The NMOS transistor has input from V_{ss} (ground) and the PMOS transistor has input from V_{dd} .
- The terminal Y is output. When a high voltage ($\sim V_{dd}$) is given at input terminal (A) of the inverter, the PMOS becomes an open circuit, and NMOS switched OFF so the output will be pulled down to V_{ss} .



CMOS Inverter



VIN	Q ₁	Q ₂	Output
0	ON	OFF	1
1	OFF	ON	0

Table 3.1 Truth table of Inverter

Questions

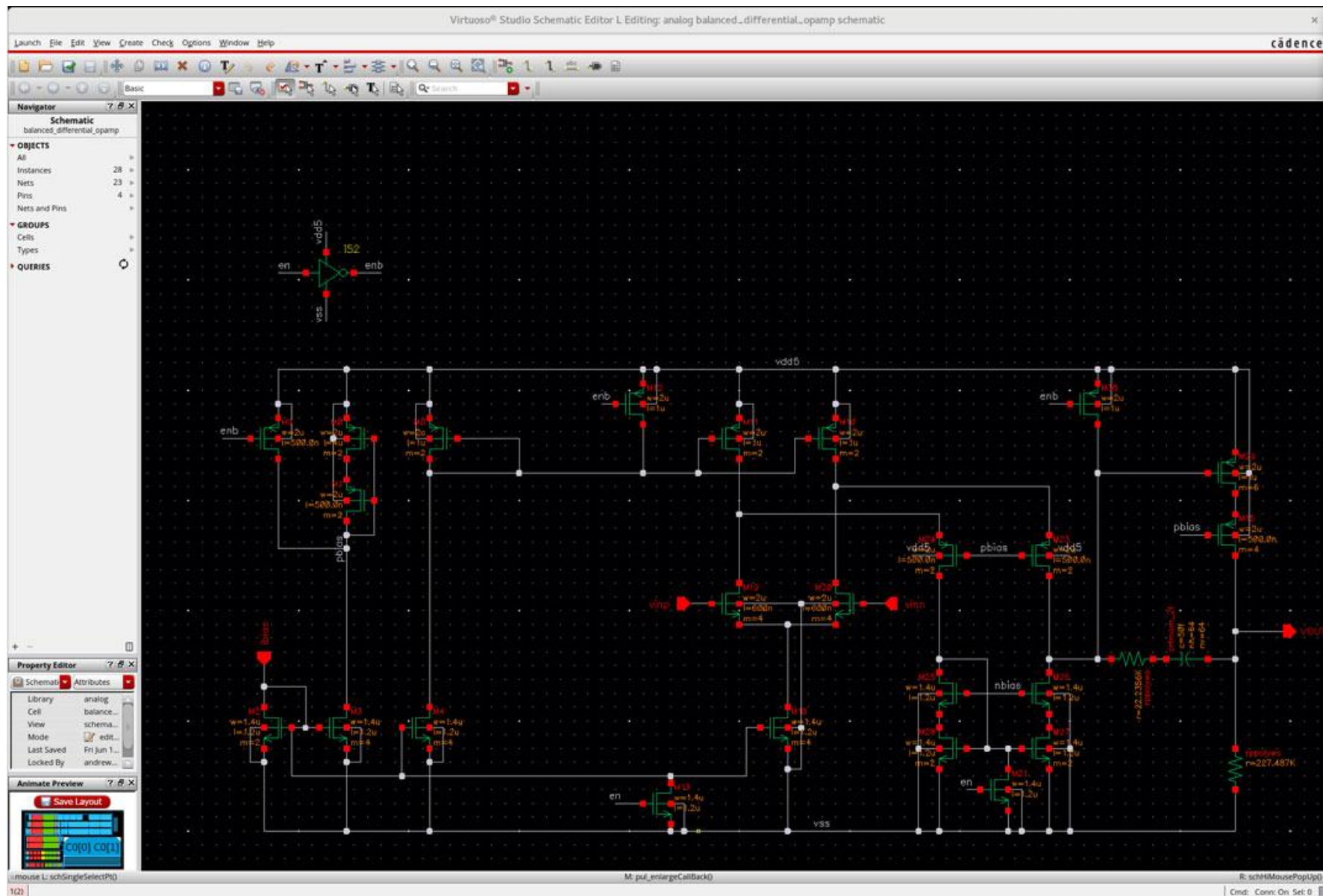
- ▶ **Why is PMOS good to pass logic 1 and NMOS is good to pass logic 0?**
- ▶ **How does a CMOS Inverter works?**
- ▶ **PMOS Vs NMOS**

Cadence Virtuoso

- Cadence Virtuoso is a **software suite** used by engineers to **design and build integrated circuits (ICs)** – the tiny “brains” inside computer chips, phones, and many electronics.
- It’s produced by Cadence Design Systems – a major company in electronic-design automation (EDA).

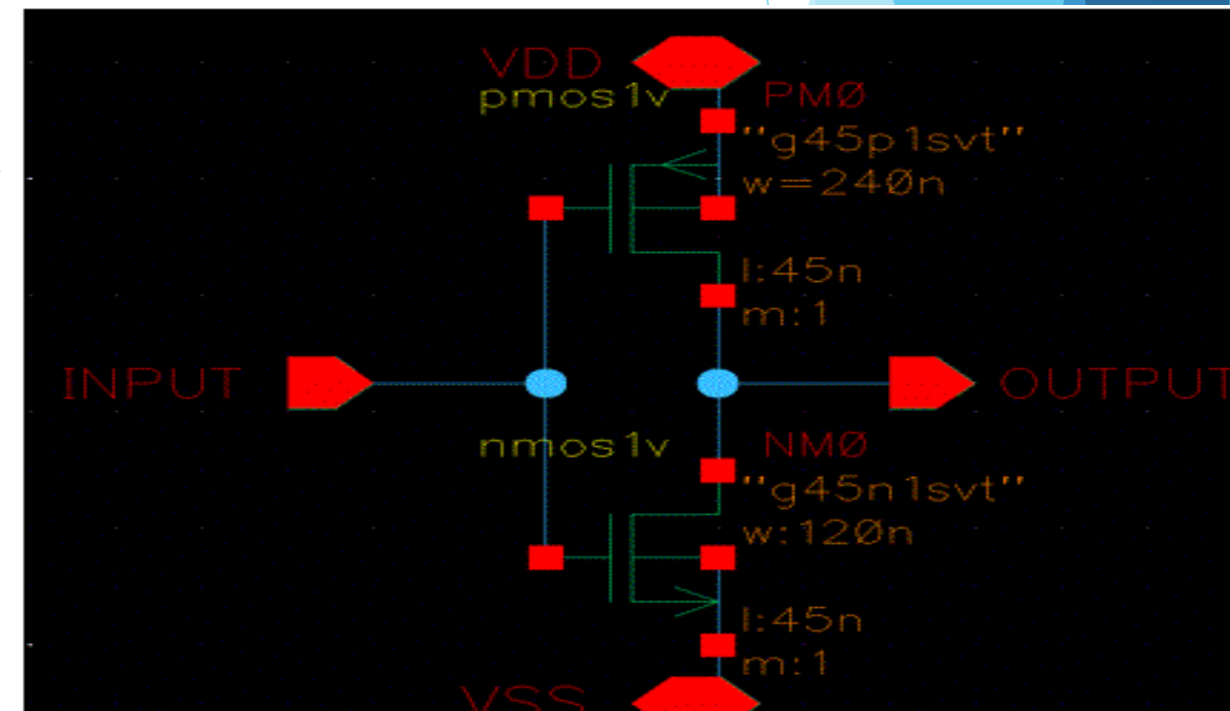
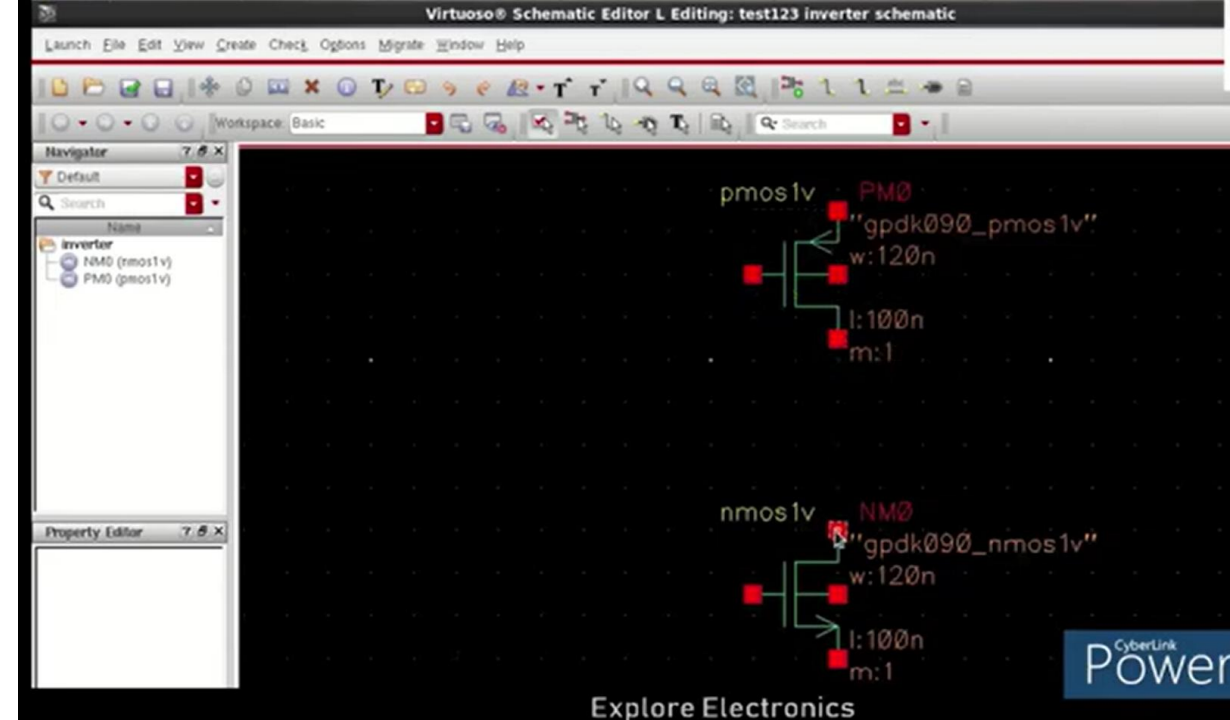
What does it do

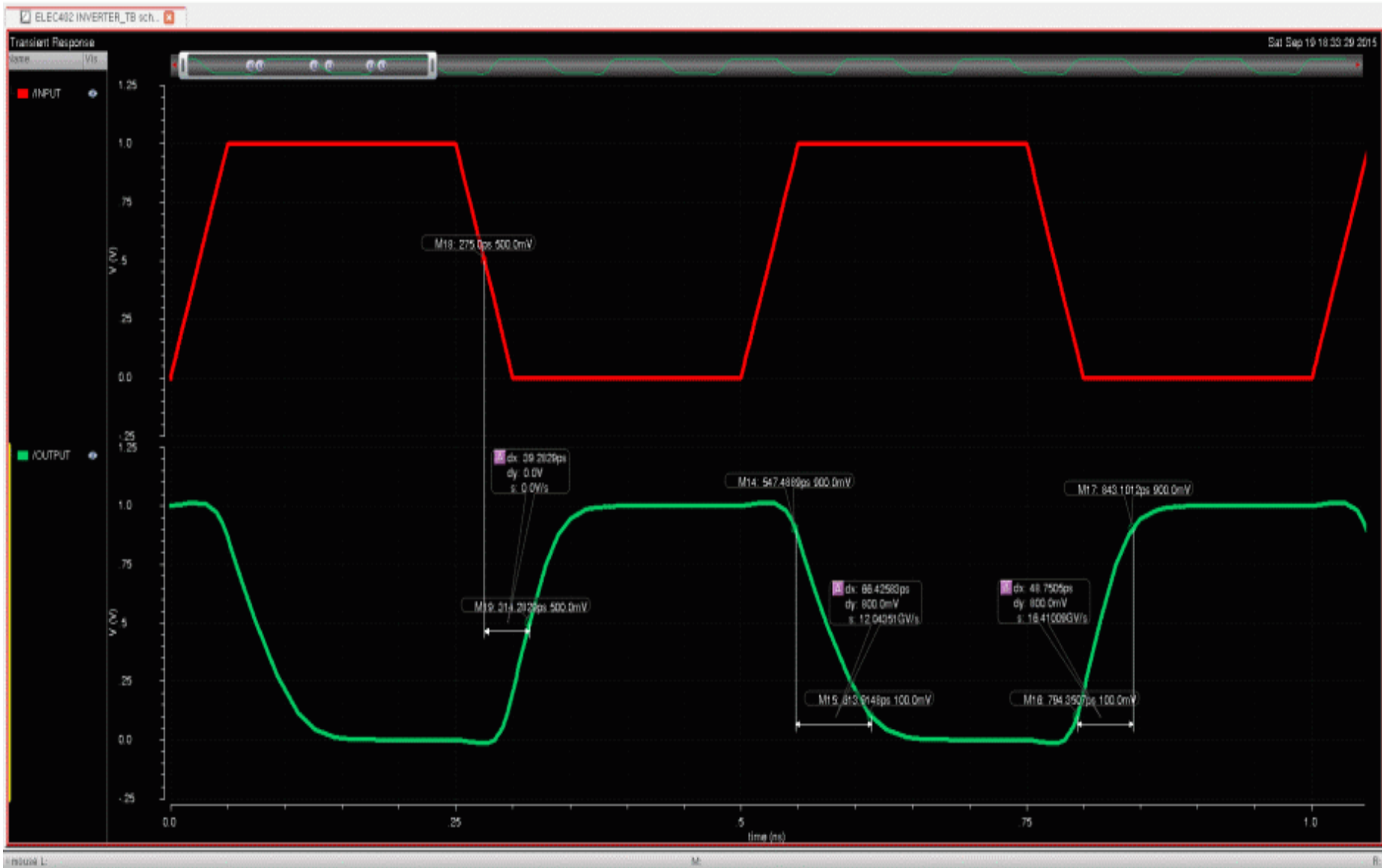
- **Schematic design** – you draw how components (transistors, resistors, etc.) connect, like a circuit blueprint using “Schematic Editor”
- **Simulation / analysis** – you can simulate how the circuit behaves (voltage, current, signal behavior, analog/RF, mixed analog-digital etc.), using parts of Virtuoso like the “ADE Suite.”
- **Layout / physical design** – once the circuit logic is set, you design how it will physically map onto silicon (or a chip), placing components, routing wires, checking for design-rules etc. That’s done with the “Virtuoso Layout Suite.”
- **Verification / full-chip integration** – Virtuoso supports advanced checks to make sure the final layout matches the schematic, that there are no errors and helps integrate.



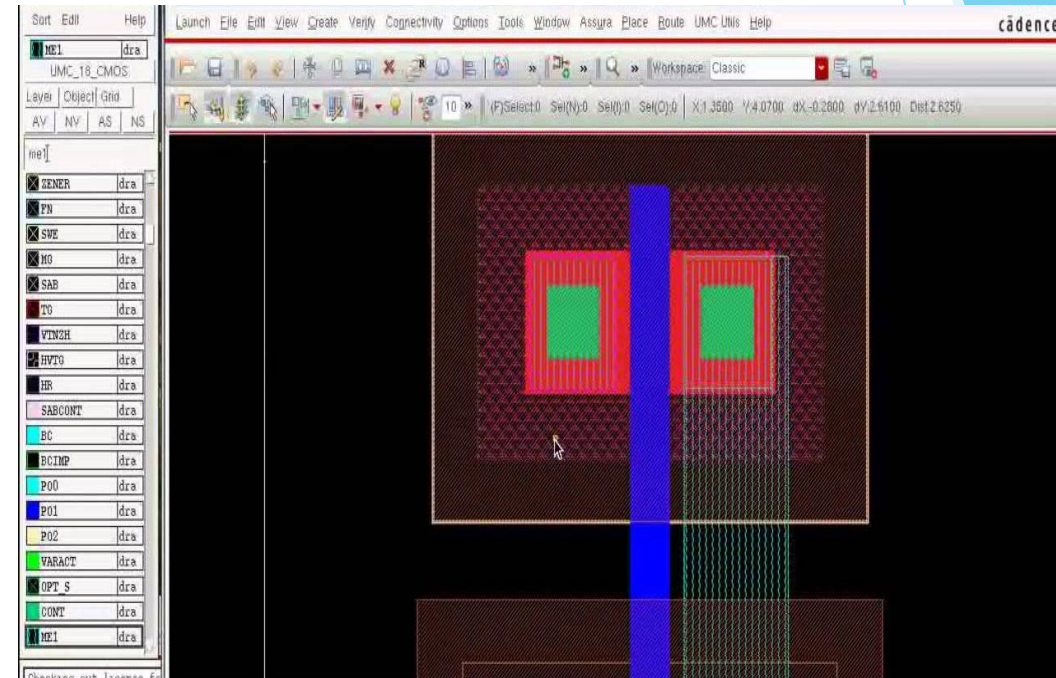
Algorithm

- ▶ 1. START
- ▶ 2. Select target semiconductor technology
 - ▶ (e.g., CMOS 180nm, 65nm, 5nm).
- ▶ 3. Load corresponding technology models + rules
 - ▶ (device models, layer rules, PDK).
- ▶ 4. Create the circuit description (schematic):
 - ▶ a. Place electronic components (transistors, resistors, capacitors).
 - ▶ b. Connect components as per desired circuit function.
- ▶ 5. Perform schematic checks:
 - ▶ a. Electrical rule check (ERC).
 - ▶ b. Correct any structural/connection errors.

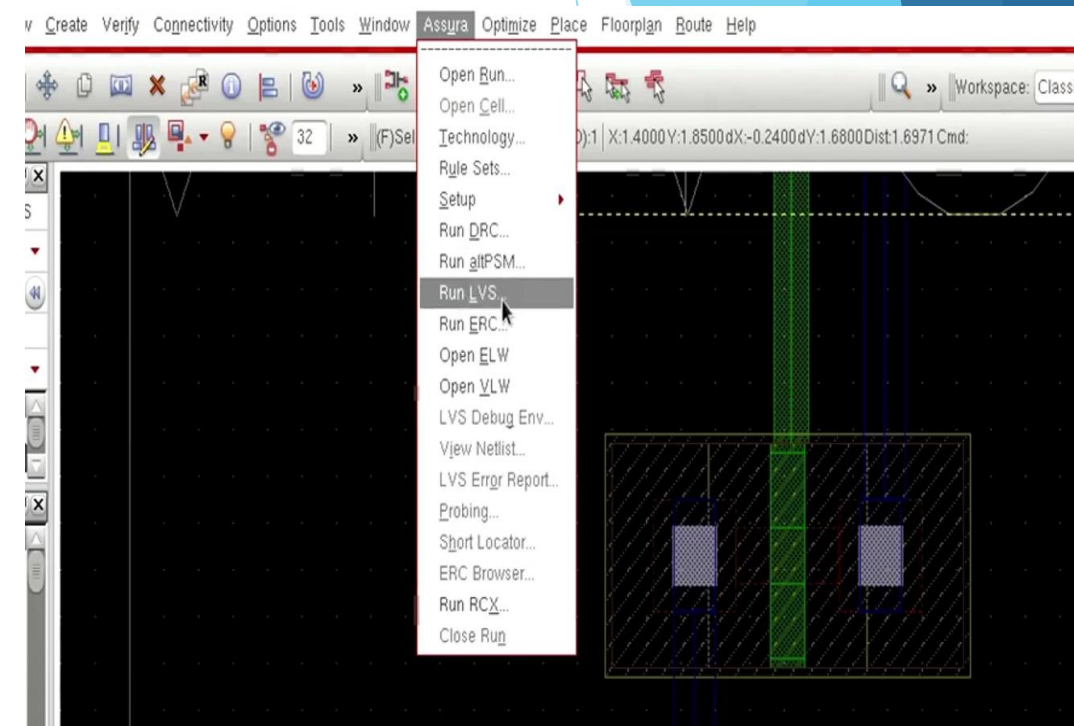




- ▶ 6. Setup circuit simulation parameters:
 - ▶ a. Choose analysis type, Set component values, signals, voltage sources & Define output measurements (voltages, currents, gain, delay).
- ▶ 7. Run circuit simulation.
- ▶ 8. If circuit does NOT meet desired performance:
 - ▶ Modify component values, sizes, or topology.
 - ▶ Go back to Step 6.
- ▶ 9. Once schematic satisfies requirements → Generate physical layout.
- ▶ 10. Create layout using design rules



- ▶ **11. Perform Design Rule Check (DRC):**
 - ▶ If violations exist → Fix layout → Repeat Step 11.
- ▶ **12. Perform Layout vs. Schematic Check (LVS):**
 - ▶ Compare layout connectivity with schematic.
 - ▶ If mismatch → Fix layout → Repeat Step 12.
- ▶ **13. Extract parasitic effects (R, C, parasitic elements) from layout**
 - ▶ to form an accurate post-layout netlist.
- ▶ **14. Perform post-layout simulation using extracted netlist.**
- ▶ **15. If performance degradation :**
 - ▶ Optimize schematic or layout as needed.
 - ▶ Go back to Step 10 or Step 4.
- ▶ **16. If design passes post-layout verification:**
 - ▶ Generate final manufacturable data (e.g., GDSII).
- ▶ **17. Send final layout output to fabrication (tape-out)**
- ▶ **18.END**



▶ <https://www.youtube.com/watch?v=e1xvh7qjzoY>